IT Support Troubleshooting Lab

Project Overview

This project demonstrates a structured troubleshooting workflow across Windows and Linux systems in a small enterprise lab environment. The lab simulates real-world incidents that an IT Support Technician or Junior System Administrator may encounter in daily operations.

The objective is to diagnose, investigate, and resolve issues involving:

- Windows client performance
- DNS resolution failures
- DHCP configuration
- Remote server support via SSH
- Basic security verification

The environment includes Windows and Linux systems connected through a firewall and internal Active Directory infrastructure.

Lab Environment

Infrastructure

- pfSense Firewall – 192.168.10.1
- Windows 11 Client A – 192.168.10.10
- Windows Server 2025 (Domain Controller / DNS / DHCP) – 192.168.10.20
- Ubuntu Server – 192.168.10.30
- Ubuntu Desktop – 192.168.10.40
- Domain Name: lab.local

Technologies Used

- Windows Server 2025
- Active Directory Domain Services
- DNS
- DHCP
- Windows 11
- Ubuntu Server
- Ubuntu Desktop
- pfSense Firewall
- SSH
- UTM Virtualization

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Troubleshooting Methodology

All incidents were investigated using a structured troubleshooting process:

1. Identify the problem
2. Gather diagnostic data
3. Analyze potential causes
4. Implement corrective actions
5. Verify system functionality
6. Document findings

Incident 1 – Windows Client Performance Issue

Problem
A Windows 11 user reported slow system performance and occasional freezes during login.

Investigation
Diagnostic tools used:

- Task Manager
- Event Viewer

Findings
System resource usage was inspected and event logs reviewed to identify potential performance bottlenecks.

Resolution
System resources and running processes were analyzed to verify that no abnormal CPU or memory consumption was present.

Incident 2 – DNS Resolution Failure

Problem
The Windows client could access internal resources but failed to resolve external domain names using DNS queries.

Example error:
DNS request timed out during **nslookup google.com**.

Investigation
Commands used:

- ipconfig /all
- nslookup google.com

The DNS server list showed the following order:

- IPv6 link-local address

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- 192.168.10.20
- 8.8.8.8

Root Cause
The Windows client prioritized an IPv6 link-local DNS server automatically advertised by the gateway. This address was not configured to provide DNS resolution.

Resolution
IPv6 was disabled on the Windows client network adapter to ensure DNS queries were directed to the internal Active Directory DNS server (192.168.10.20).

Incident 3 – DHCP Service Verification

Problem
Clients were suspected to have inconsistent network configuration behavior.

Investigation
The DHCP service on the Windows Server was inspected using the DHCP management console.

Lease information and scope configuration were reviewed to verify correct address assignment.

Resolution
DHCP functionality was confirmed to be operational after service verification.

Incident 4 – Remote Server Support (SSH)

Problem
A remote user required assistance accessing a service hosted on an Ubuntu server.

Investigation
The SSH service was inspected using:
- systemctl status ssh

Connection testing was performed from the Windows client using an SSH session.

Resolution
The SSH service was confirmed active and running. Remote connectivity to the Ubuntu server was successfully established.

Security Verification

Basic security validation was performed after troubleshooting to confirm system protection mechanisms were active.

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Checks included:

- Windows Defender Firewall status
- System update verification
- Network connectivity validation

Final System Verification

The environment was validated using connectivity and DNS tests:

- ping 192.168.10.1
- ping 192.168.10.20
- ping google.com
- nslookup google.com

All tests returned successful results confirming full network and DNS functionality.

Key Skills Demonstrated

1. Cross-platform troubleshooting (Windows & Linux)
2. DNS diagnostics and configuration
3. Network connectivity testing
4. Service validation
5. Remote administration via SSH
6. Structured troubleshooting methodology

Conclusion

This lab demonstrates practical troubleshooting techniques across multiple systems and services within a small enterprise network environment. The investigation process highlights the importance of structured diagnostics, accurate root cause analysis, and verification of corrective actions.

The project reflects the type of operational tasks performed by IT Support Technicians and Junior System Administrators responsible for maintaining reliable system and network infrastructure.

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